1. n letters

n envelopes

L1 L2 ... Ln-1 Ln

↓

E1 E2 ... En-1 En

Sample Space $(n!)$

2. Elevator 5 passengers & 7 floors

Pass 1 → 1 ... 7

Pass 2 → 〃

⋮

Pass 5

Sample 7^5

↓

$7 \times 6 \times 5 \times 4 \times 3$

$$\frac{7 \times 6 \times 5 \times 4 \times 3}{7^5}$$

3.

G: Genius C: likes chocolate

$P(G) = 0.6$, $P(C) = 0.7$, $P(G \cap C) = 0.4$ $\quad P(G^c \cap C^c) = ?$

$P(G \cup C) = P(G) + P(C) - P(G \cap C)$

$0.6 + 0.7 - 0.4 = 0.9$

$P(G^c \cap C^c) = 1 - P(G \cup C) = 1 - 0.9 = 0.1$

$2 \to 0$

$3 \to \frac{4}{51}$

$4 \to \frac{8}{51}$

$\vdots$

$Ace \to \frac{48}{51}$

$$\frac{1}{13}\left[0 + \frac{4}{51} \cdots \frac{48}{51}\right]$$

$$\Rightarrow \frac{1}{13} \times \frac{4}{51}\left[1 + 2 \cdots 12\right]$$

$$\Rightarrow \frac{4^2}{13 \times 51} \times \frac{(12)(13)}{2} = \frac{24}{51} = \frac{8}{17}$$

E_1 = You choose larger E_2 : Your friend chooses larger E_3: You choose the same

$P(E_1) = P(E_2)$

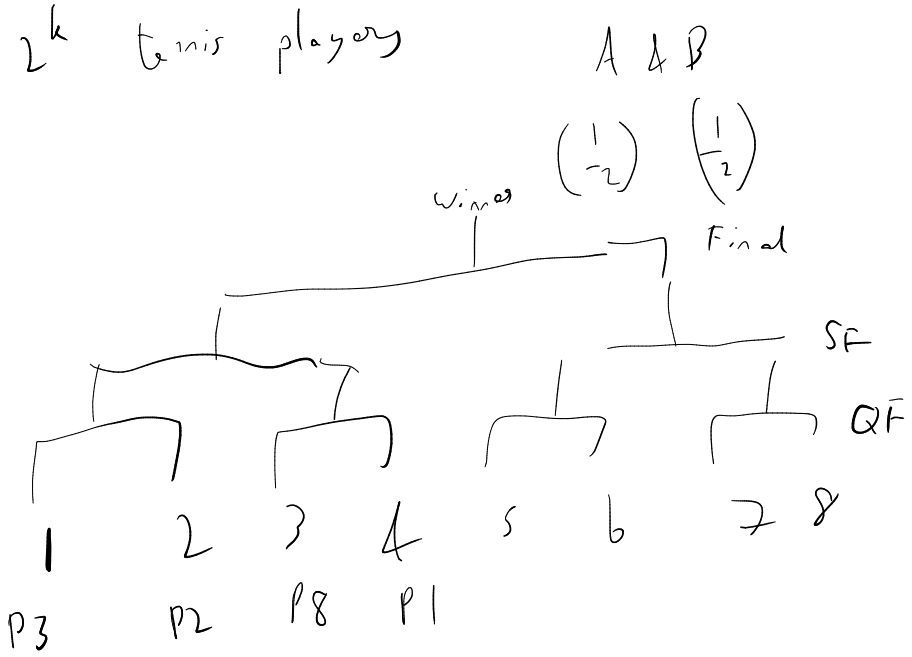
$P(E_1) + P(E_2) + P(E_3) = 1$

$2P(E_1) + P(E_3) = 1$

$P(E_3) = \frac{3}{51} = \frac{1}{17}$

$2P(E_1) = \frac{16}{17} \Rightarrow P(E_1) = \frac{8}{17}$

b) 2^k tennis players



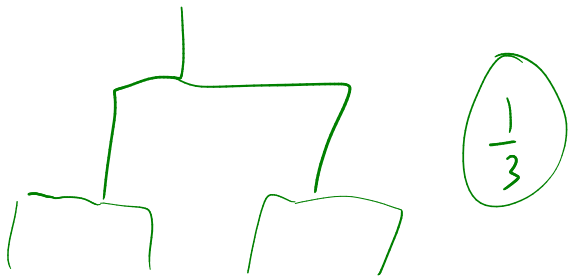
i. P(P1 & P2 meet in the First round)

Fava → 2^k

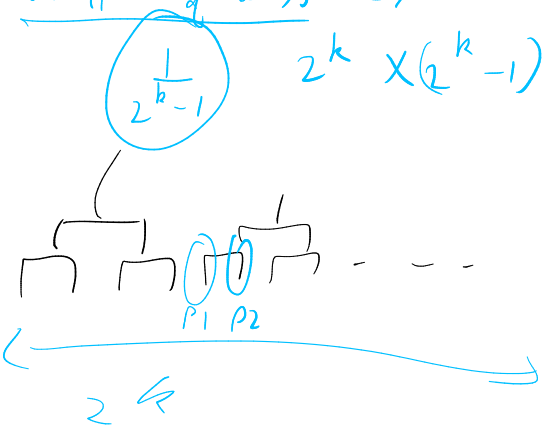
Total # of ways →

$2^k \times (2^k - 1)$

$\frac{1}{2^k - 1}$



$\frac{1}{3}$



Alcaraz ✓

Djokovic

Sinner ✓

Fritz

$P(P1 \& P2 \text{ meet in the finals}) = \frac{2}{2^k \cdot (2^k - 1)}$

P1 & P2, P1 & P3, P2 & P3, P1 & P4, P2 & P4, P3 & P4

$^{2^k}C_2$

3. p_n {P1 & P2 have a match}

of games in a tournament → $2^k - 1$

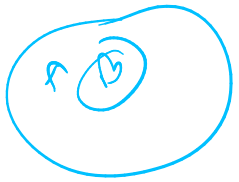
of possible player pairs → $\frac{2^k \cdot (2^k - 1)}{2}$

$\frac{2^k - 1}{\frac{2^k \cdot (2^k - 1)}{2}}$

$P(A) \geq P(A \cap B)$

GLB

LUB



$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\cup B \leq P(B)$

$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$\cup B \geq P(A) + P(B) - 1$